

DANIELA AROS MUALIN

Postdoctoral researcher

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Research profile

My research focuses on *ecophysiology and ecology of seed-free plants*. I study how plants adapt to different environments, covering from functional traits along environmental gradients to more detailed studies on gas exchange and circadian rhythms among different climates and phylogenetical clades. I have focused primarily on ferns because they can provide different insights into land plant evolution from the more common model plants and have been severely understudied.

Education

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| 2020–2024 | Ph.D. Ecology , University of Zurich, Switzerland.
Dissertation: Carbon- and water-balance among ferns. Faculty mentor: PD. Michael Kessler |
| 2015–2017 | M.Sc. Plant Science , University of Bonn, Germany.
Dissertation: Functional diversity in ferns is driven by species richness rather than environmental constraints. Faculty mentor: Prof. Dietmar Quandt & PD. Michael Kessler. |
| 2011–2014 | B.Sc. Biology , Pontificia Universidad Católica de Chile, Chile. |

Professional experience

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| 2024–present | Postdoctoral researcher , Purdue University, USA |
| 2018 | Technical assistant , Pontificia Universidad Católica de Chile, Chile.
DNA extraction and microsatellite amplification. Prof. Maureen Murúa FONDECYT 11170377 |
| 2018 | Field work manager , University of Zurich, Switzerland. |
| 2016–2017 | Botanical Garden touristic guide , University of Bonn, Germany. |
| 2015 | Lab manager , Pontificia Universidad Católica de Chile, Chile. Prof. Fernanda Pérez. |
| 2014 | Technical assistant , Pontificia Universidad Católica de Chile, Chile.
DNA extraction and microsatellite amplification. Prof. Fernanda Pérez FONDECYT 1141047 |

Grants, awards & memberships

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| 2026–2030 | DFG Research Unit (GArua): Fog-dependent water relations and plant invasion in the Galápagos Archipelago under climate change (accepted, |
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	pending final budget approval). Principal Investigator. Subproject: Epiphyte interactions with climate in the Galápagos humid zone – 400.000 EUR
2024–2026	Swiss National Science Foundation Postdoctoral Mobility grant. P500PB_221982/1 Title: Exploring the evolution and function of Crassulacean Acid Metabolism (CAM) in ferns -120.000 CHF
2023	GRC Travel Grant, Graduate Campus, University of Zurich - 3500 CHF
2022	Georges and Antoine Claraz-Schenkung (Foundation)– 3000 CHF
2022	Partial Scholarship, Organization of Tropical Studies – 3000 USD
2021	Georges and Antoine Claraz-Schenkung (Foundation)– 3000 CHF
Member	American Fern Society

Publications

Peer-reviewed

1. Kessler M & **Aros-Mualin D** (in press). Fern and lycophyte ecology: from physiology to communities. In: Zhang LB (ed.), *The Families and Genera of Ferns and Lycophytes, Vol. I: Biology, Evolution, and Classification*. Springer (update of Kubitzki's *The Families and Genera of Vascular Plants*).
2. **Aros-Mualin D** & McAdam SAM (2026, in press). The dawn of stomatal control by the hormone of darkness. *Journal of Experimental Botany*, erag243. <https://doi.org/10.1093/jxb/erag243>
3. Páez S, **Aros-Mualin D**, Kessler M, Kluge J & Meza-Torres EI (2026). The refuge effect of humid microhabitats for ferns decreases towards more arid regions. *Frontiers in Ecology and Evolution*, 14, 1719536. <https://doi.org/10.3389/fevo.2026.1719536>
4. McAdam SAM, Manandhar A, Rimer IM & **Aros-Mualin D** (2025). Exclusive and hypersensitive stomatal control by blue-light in *Equisetum*. *Plant Physiology*, kiaf238. <https://doi.org/10.1093/plphys/kiaf238>
5. **Aros-Mualin D** & Kessler M (2024). Untangling poikilohydry and desiccation tolerance: Evolutionary and macroecological drivers in ferns. *Annals of Botany*, 134(7), 1139–1150. <https://doi.org/10.1093/aob/mcae167>
6. Kessler M & **Aros-Mualin D** (2024). The power of independent generations in plants. *New Phytologist*, 2(245), 440–442. <https://doi.org/10.1111/nph.20162>
7. Neubauer N, **Aros-Mualin D**, Mariscal V & Szövényi P (2024). Challenging the term symbiosis in plant–microbe associations to create an understanding across sciences. *Journal of integrative plant biology*, 66 (1), 7-11. <https://doi.org/10.1111/jipb.13588>
8. **Aros-Mualin D**, Flores-Galván C, Páez S, Pelosi JA, Sorojsrisom E, Yawn N & Watkins Jr. JE (2023). In Situ Observations of the Gametophytes of Six Fern Species of Costa Rica. *American Fern Journal*, 113 (3), 170-190. <https://doi.org/10.1640/0002-8444-113.3.170>
9. **Aros-Mualin D**, Guadagno CR, Silvestro D & Kessler M (2023). Light, rather than circadian rhythm, regulates gas exchange in ferns and lycophytes. *Plant Physiology*, kiad036. <https://doi.org/10.1093/plphys/kiad036>
10. **Aros-Mualin D**, Flexas J, Galbier F & Kessler M (2022). Exploring the Ecological Relevance and Variability of Circadian Regulation in Marsileaceae. *American Fern Journal*, 112(4), 303–319. <https://doi.org/10.1640/0002-8444-112.4.303>

11. Ossa CG, **Aros-Mualin D**, Mujica MI & Pérez F (2021). The physiological effect of a holoparasite over a cactus along an environmental gradient. *Frontiers in plant science*, 12, 763446. <https://doi.org/10.3389/fpls.2021.763446>
12. **Aros-Mualin D**, Noben S, Karger DN, Carvajal-Hernandez CI, Salazar L, Kluge J, Lehnert M, Quandt M & Kessler M (2021). Functional diversity in ferns is driven by species richness rather than by environmental constraints. *Frontiers in plant science*, 11. <https://doi.org/10.3389/fpls.2020.615723>
13. Weigand A, Abrahamczyk S, Aubin I, Bitá-Nicolae C, Bruehlheide H, Carvajal-Hernández C, ... **Aros-Mualin D**, ... & Kessler M (2020). Global fern and lycophyte richness explained: How regional and local factors shape plot richness. *Journal of Biogeography*, 47(1), 59-71. <https://doi.org/10.1111/jbi.13782>

In Review

1. **Aros-Mualin D**, Manandhar A & McAdam SAM. VPD and C_i drives inverted stomatal rhythms in an obligate CAM fern. *New Phytologist*, in revision.
2. **Aros-Mualin D**, Wüthrich S, Kessler M, Suissa JS. The evolution and ecology of fertile-sterile dimorphism in ferns. *New Phytologist*, under review.
3. **Aros-Mualin D**, Brodribb T, McAdam SAM. The origin and evolution of stomatal function. Invited review, *Annals of Botany*, in preparation.
4. **Aros-Mualin D**, McAdam SAM, Manandhar A & Kessler M. Can gas exchange be predicted from stomatal traits? In preparation.

Teaching experience

Teaching assistant

2020–2023	Biodiversity and biogeography . University of Zurich. Instructor: PD. Michael Kessler. Intensive course on project design and grant writing in biogeography. Led writing workshops and invited lectures.
2022-2023	Evolution and biodiversity . University of Zurich. Instructor: Prof. Florian Schiestl, PD Dr. Thomas Posch, and Prof. Marcel G. A. van der Heijden. Basic undergraduate course. Led a few laboratory sessions dedicated to seed-free plants.
2014–2013	Plant Biology and biodiversity . Pontificia Universidad Católica de Chile. Instructor: Prof. Fernana Perez. Led laboratory sessions and field excursions.
2014	Ecology . Pontificia Universidad Católica de Chile. Instructor: Prof. Pablo Marquet. Led workshops that complemented and reinforced course lectures.
2014	Plant physiology . Pontificia Universidad Católica de Chile. Instructor: Prof. Fernana Perez. Led laboratory sessions that complemented and reinforced course lectures.

Mentoring

2026–ongoing	Daniela Araneda Orellana, P.hD. Ecology, Pontificia Universidad Católica de Chile
2026	Hannah Frances Stratton, Undergraduate, Purdue University.

2026	Abigail Carr, Undergraduate, Purdue University.
2026	Caleb Anthony Judy, Undergraduate, Purdue University.
2025-2026	Giulia Maddalena, Undergraduate, University of Zurich.
2025	Julia Peterson, Undergraduate, Purdue University.
2023–2024	Roman Gmür, M.Sc. Biology, University of Zurich.
2020–2021	Sarina Wüthrich, M.Sc. Biology, University of Zurich.

Presentations

Invited talks

2022	Invited speaker at Plant Ecology group seminar, Institute for Evolution and Ecology, Tübingen University, Tübingen, Germany. Title: “Water-carbon relations in ferns”
2020	Invited speaker at IBONE Seminar, Institute of Botany of the Northeast, National University of the Northeast, Corrientes, Argentina (in Spanish). Title: “La evolución de la relación carbono y agua en Helechos y Licófitas”

Conference presentations

1. Aros-Mualin D* & McAdam SAM (2026). Unique stomatal responses of a CAM fern to changes in CO₂ concentration in the dark. *XIX Summer colloquium on Plant Ecophysiology*, Katalapi, Chile. (presentation)
2. Aros-Mualin D*, Manandhar A & McAdam SAM (2025). Do CAM ferns have inverted stomatal rhythms?. *Botany 2025–Botany without borders*, Palm Springs, USA. (presentation).
3. Aros-Mualin D*, McAdam SAM (2025). Como invierten el ritmo los estomas de helechos CAM?. *XII Coloquio de primavera sobre Ecofisiología Vegetal*, Mallorca, Spain. (presentation).
4. Aros-Mualin D* & McAdam SAM (2025). Do CAM ferns have inverted stomatal rhythms?. *6th Xylem International Meeting*, Sevilla, Spain. (poster).
5. Aros-Mualin D* & McAdam SAM (2025). Do CAM ferns really exist?. *XVIII Summer colloquium on Plant Ecophysiology*, Katalapi, Chile. (presentation)
6. Aros-Mualin D & Kessler M (2024). Water use efficiency explain differences in global diversity patterns between ferns and angiosperms. *XX International botanical congress*, Madrid, Spain. (presentation)
7. Aros-Mualin D* & Kessler M (2023). El agua como una limitación clave en los helechos. *XI Coloquio de primavera sobre Ecofisiología Vegetal*, Navarra, Spain. (presentation)
8. Aros-Mualin D* & Kessler M (2023). Water as a key limitation for ferns. *XVI Summer colloquium on Plant Ecophysiology*, Katalapi, Chile. (presentation)
9. Aros-Mualin D* & Kessler M (2022). Why are ferns more desperate for water than other plants. *Annual Meeting of the Ecological Society of America (ESA) and Canadian Society of Ecology and Evolution (CSEE)*, Montreal, Canada. (presentation)
10. Aros-Mualin D* & Kessler M (2020). Evolution of carbon and water relations among lycophytes and ferns. *XV Summer colloquium on Plant Ecophysiology*, Katalapi, Chile. (presentation)

11. Aros-Mualin D* & Kessler M (2018). Functional diversity in ferns is driven by species richness rather than by environmental constraints. *Annual Meeting of the Ecological Society of Germany, Austria and Switzerland (GfÖ)*. Vienna, Austria. (presentation)
12. Aros-Mualin D*, Ossa CG & Perez F (2014). *Echinopsis chiloensis* does not compensate for the sink effect of the holoparasite *Tristerix aphyllus*. *MEDECOS XIII International Conference*, Olmue, Chile. (presentation)
13. Aros-Mualin D*, Ossa CG & Perez F (2014). El holoparasito *Tristerix aphyllus* ¿Afecta la adecuación biológica o fisiológica de su principal hospedero *Echinopsis chiloensis*?. *XI Congreso Latinoamericano de Botánica*, Salvador de Bahía, Brazil. (poster)
14. Aros D*, Ossa CG, Campano F, Bull-Hereñu K, Hinojosa F & Perez F (2013). Xylem functional traits in *Myrceugenia* contradicts evolutionary theory of habitat adaptation. *V Binational Ecology Congress of Chile and Argentina*, Puerto Varas, Chile. (poster)

Professional services

2023–2024	Member of Hiring Committee for Assistant Professorship Tenure Track in Plant Diversity and Macroevolution, University of Zurich, Switzerland (Nov 2023–Jun 2024) • Served as the PhD student representative on a 14-member committee to hire a new Assistant Professor. • Participated in candidate evaluation, interviews, and final hiring decisions, representing the perspectives of early-career researchers.
2021-present	Peer Reviewer for Scientific Journals: regularly contribute to the peer review process for journals <i>New Phytologist</i>, <i>Scientific Reports</i>, <i>Physiologia Plantarum</i>, <i>Plant, Cell & Environment</i>, <i>Biotropica</i>, <i>Plant Physiology</i>, and <i>American Fern Journal</i>. My reviews focus on topics in plant physiology and ecology.